

EARTHPRESS · CALCULATION NOTE

EarthPress sizing calculations

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.3	2026-10-01	Amish Chadha	Constructable design (EPR-DDR-003): lid span 420 mm between the hinge and latch pins, link net section at the 41 mm bush, column access holes, eject arm, mass with guard and bolts (200 kg, R7 not met), cost \$520 (R13 not met)
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). R5, R7 and R13 checked against the accepted targets (500 N per operator, grip 0.8 to 1.9 m with the peak at 1.0 m or higher; 190 kg total; \$500 budget); script reads the budget from project.yaml
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing note (force, linkage kinematics, grip force and work, fill tolerance, structure at the design load, ejection, mass, output, material flow, carbon, cost) against every requirement in EPR-REQ-001

EarthPress sizing calculations

On paper the press reaches 2 MPa on the 290 x 140 x 90 mm block, but only with **two people on the lever** and a **weighed fill**. The toggle and 1.7 m lever give a force ratio of 173:1 at the stop, and the peak pull under the base soil assumption is **668 N**, which comes about 3 mm before the end of the stroke. One operator pulling 500 N stalls at about 0.5 MPa, so the press is a two-person machine, as Amish decided on 2026-09-25 (EPR-DDR-002). Against the targets he accepted that day, **R5** is met on paper (334 N per operator, grip 0.86 to 1.89 m, peak at 1.16 m). Since the design was made buildable on 2026-10-01 (EPR-DDR-003, v0.3 of this note), **R7 is not met** (200 kg against 190 kg; no piece over 45 kg, so the 50 kg piece limit is met) and **R13 is over its value-engineering target** (estimated \$520 against a \$500 target, USD 20 over); R7 is an open decision and the cost savings worth trying are in the design decisions register (EPR-DEC-001). R2 and R4 are **at risk**; R3, R8, R10 and R14 cannot be verified at TRL 3.

The calculation corrects four TRL 2 figures and one TRL 2 mechanism. The press mass rises from about 123 kg to 187 kg, the cost from about \$403 to \$488 and the compaction work falls from 0.6 to 1.2 kJ to about 0.35 to 0.71 kJ (0.49 kJ base). The 22 mm lid would reach yield at 2 MPa and is now a ribbed lid. The TRL 2 concept ejected the block by pulling the lever on past the compaction point, which a toggle ending near straight cannot do; ejection now uses a separate seesaw lever and a lost-motion slot in the push rod.

Every number here is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`), which also writes `docs/04-calcs/results.csv`. The script reads the geometry from `cad/src/model.py` and the costs from `bom/bom.csv`. All values are first-principles estimates; nothing is measured.

■ 1. Assumptions

Table 1. Inputs. All are assumptions for a paper design.

Input	Value	Basis
Block	290 x 140 x 90 mm; loose fill 150 mm deep	EPR-DDR-001 item 3
Target pressure	2.0 MPa at the end of the stroke	EPR-DDR-001 item 4, R2
Dry block density	1,900 kg/m ³	TRL 2 assumption
Stabilizer and water	5 % cement on dry soil, 10 % water on dry mix	EPR-DDR-001 item 5
Soil compaction law	Pressure rises tenfold over the last 14 mm of the stroke (exponential); range 10 to 20 mm	Assumed; to be measured with partner soil
Operators	500 N for one person (R5 limit); 1,000 N for two on the T-handle, also the abuse case	Assumed
Working grip band	0.8 to 1.9 m above ground, peak force at 1.0 m or higher	R5 as accepted by Amish, EPR-DDR-002
Steel	Yield 250 MPa (plate and sections), 240 MPa (A53 grade B pipe), 370 MPa (C45 pins); E = 200 GPa	Lower-bound catalog values
Bush bearing pressure	100 MPa allowable, hardened steel, greased, slow oscillation	Assumed
Lateral pressure in the mold	0.5 x vertical during pressing; 0.1 MPa residual after unloading, friction 0.6	Assumed
Soil rebound at 2 MPa	1.0 mm	Assumed
Bolts	M16 8.8, 60.3 kN each in single shear through the thread	$0.6 \times 800 \times 157 / 1.25$
Fatigue	FAT 71 welded detail, 243 MPa range at 50,000 cycles	$71 \times (2 \times 10^6 / 5 \times 10^4)^{(1/3)}$
Cycle	85 s per block (Table 6); 7 working hours per 8 h day	Assumed, to be timed
Soil preparation	15 % oversize; sieving 400 kg/h and mixing 600 kg/h per person; 3 % rejects remixed	Assumed
Carbon	49 kg CO ₂ /m ³ for 5 % CSEB, 643 kg CO ₂ /m ³ for fired brick	Auroville Earth Institute
Steel price	About \$1.30/kg	Indicative, TRL 2 basis

■ 2. Block and force (R1, R2)

The block face is 0.0406 m², so **2 MPa needs 81.2 kN** (4 MPa would need 162.4 kN). The block holds 3.65 L and weighs 6.94 kg dry and 7.64 kg moist at the press, with 0.33 kg of cement. A 150 mm loose fill compacts to 90 mm, a ratio of 1.67, which implies a loose dry density of about 1,140 kg/m³ in the mold.

■ 3. Linkage and lever (R2, R5)

The mechanism is a bottom piston driven by a symmetric toggle (two 245 mm links), whose knee is pushed by a 420 mm connecting link from a 107 mm crank on the lever hub. The geometry in `cad/src/model.py` came from a design search that minimized the peak grip force while keeping the grip between 0.8 and 1.9 m; the script checks it.

Table 2. Kinematics.

Quantity	Value
Toggle angle from vertical, start to stop	29.3° to 6.0°
Piston stroke	60.0 mm
Base pin and upper pin heights at the stop	298 mm and 785 mm (mold rim at 940 mm)
Lever pivot	x = -468 mm, z = 266 mm; grip radius 1,650 mm
Lever angle, start to stop	99.6° to 158.8° from horizontal (+X), an arc of 59.2°
Grip path	1.71 m
Grip height, start to stop	1,893 mm to 863 mm
Force ratio at 30 mm and at the stop	20.0:1 and 173:1 ; it rises monotonically to the stop

The original R5 band could not hold a useful arc. A grip 1.65 m from the pivot can stay between 0.9 and 1.6 m for only 24.5° of arc. The TRL 2 recommendation to raise the pivot to about 800 mm for a 100° arc does not work with a lever of this length: a symmetric 100° arc would take the grip to -464 mm, below the ground. The design therefore keeps a low pivot and starts the lever nearly upright, which gives 59° of arc with the grip between 0.86 and 1.89 m. This meets the intent of EPR-DDR-001 item 2 (a longer arc and a higher grip at the end) but not its wording, and it did not meet the original 0.9 to 1.6 m R5 band. Amish accepted the reworded R5 on 2026-09-25 (grip 0.8 to 1.9 m with the peak force at 1.0 m or higher; EPR-DDR-002), and the design meets it: the grip stays inside the band and the peak comes at 1,157 mm.

■ 4. Grip force and work (R2, R5)

Table 3. Grip force through the stroke, base soil (14 mm per decade).

Travel (mm)	Piston force (kN)	Force ratio	Grip force (N)	Grip height (mm)
0	0.00	14.8	0	1,893
30	0.59	20.0	29	1,735
40	3.04	25.0	122	1,621
50	15.7	37.5	417	1,429
55	35.7	56.5	632	1,257
57	49.8	74.5	668	1,150
59	68.7	115.8	593	994
60	81.2	173.0	469	863

- **Peak grip force: 668 N** at 56.9 mm of travel, with the grip at 1.16 m (waist to chest height). With a stiffer soil (10 mm per decade) the peak is 571 N; with a softer one (20 mm) it is 809 N.
- **One operator at 500 N stalls at 51.9 mm, at 0.53 MPa**, and no overfill helps (best 0.53 MPa). The peak comes before the stop, so the operator cannot "push through" it.
- **Two operators on the T-handle** (334 N each at the peak) reach the stop at 2.00 MPa.

- Compaction work is **494 J** for the base soil (353 J and 705 J for the range). One operator at 500 N could give 853 J over the arc, so energy is not the limit; the mismatch between the force ratio and the soil curve is. A perfect match would need 58 % of the operator's capacity.

R2 is at risk (met with two operators, within the fill window of section 5, for the base soil). With the accepted R5 (500 N or less per operator, EPR-DDR-002), **R5 is met on paper** with two operators at 334 N each and the peak at 1,157 mm; one operator alone cannot make a 2 MPa block.

■ 5. Fill tolerance (R1, R2)

The crank stop sits at the design end angle (6°), so the stop limits the force the linkage can deliver (section 6) and every block that reaches the stop is 90.0 mm high. The price is a narrow fill window. If the fill mass is e above nominal, the soil reaches 2 MPa at a height of $90(1 + e)$ mm.

Table 4. Pressure and block height against fill error, two operators (1,000 N).

Fill error	Pressure	Block height
-2 %	1.49 MPa	90.0 mm
-1 %	1.72 MPa	90.0 mm
0 %	2.00 MPa	90.0 mm
+2 %	2.69 MPa	90.0 mm
+4 %	1.15 MPa (operators stall)	97.0 mm

The fill must be between **0 and +2.7 %** of nominal, that is, 0 to 206 g over 7.64 kg of moist mix. Filling by struck volume alone cannot hold this, since loose density varies with moisture and handling. The proposal is to weigh each fill to about $+1.3 \% \pm 100$ g on a 10 kg hanging scale with 50 g divisions (now in the soil test kit, BOM item 9). R1 (height 90 ± 3 mm) is met on paper within this window.

■ 6. Design load and structure (R8, R12)

Design load. The largest force the linkage can deliver is the operators' pull times the force ratio at the stop: $1,000 \text{ N} \times 173 = \mathbf{173 \text{ kN}}$ (4.26 MPa on the block). All members are checked at this abuse load and at the nominal 81.2 kN. It exceeds the R12 latch requirement of $1.5 \times 81.2 = 121.8 \text{ kN}$.

Table 5. Structural checks (safety factor on yield or capacity).

Member	Nominal 81.2 kN	Abuse 173 kN
Toggle pins, 35 mm C45, double shear	42 MPa (5.03)	90 MPa (2.36)
Bush bearing, 2 x 28 mm links	42 MPa (2.40)	89 MPa (1.13)
Link net section at the 41 mm bush hole, 28 x 70 mm	50 MPa (4.97)	107 MPa (2.33)
Link buckling out of plane	SF 103	SF 48
Push rod net section at the slot, 60 mm square	59 MPa (4.25)	125 MPa (1.99)
Lid, 20 mm plate with two 20 x 70 mm ribs, 420 mm between hinge and latch pins	73 MPa (3.43)	155 MPa (1.61)
Lid deflection at mid-span	0.08 mm	0.17 mm
Hinge and latch pins, 30 mm, double shear	29 MPa (7.43)	61 MPa (3.49)
Mold long wall with the 16 x 50 mm belt	89 MPa (2.80)	190 MPa (1.31)
Columns, UPN 80 in tension	37 MPa (6.77)	79 MPa (3.18)
Columns at the 40 mm pin access hole	47 MPa (5.30)	101 MPa (2.49)
Mold to column bolts, 8 x M16 8.8	SF 5.9	SF 2.8
Base beam, two 20 x 100 mm plates	73 MPa (3.41)	156 MPa (1.60)
Connecting link, 50 x 24 mm, 42.2 kN at the abuse case		buckling SF 15.3
Lever, 2 in schedule 40 at the socket mouth	102 MPa at the 668 N peak (2.35)	153 MPa at 1,000 N (1.57)

The lowest factor is 1.13 (bush bearing at the abuse load). The connecting link acts 39.1 mm from the lever pivot at the stop and carries 42.2 kN at the abuse case; the two tie tubes from the lever bracket to the base beam carry a horizontal 34.1 kN. The mold wall bulges 0.08 mm at 2 MPa and the columns stretch 0.16 mm, so neither affects block size.

The TRL 2 lid does not work. A plain 22 mm plate, 220 mm wide, spanning the 420 mm between hinge and latch pins reaches 315 MPa at the nominal 81 kN, above yield. The ribbed lid in the model is the fix; it weighs 27 kg with its pins, which the operator lifts at the latch end every cycle (about 133 N).

Fatigue (R8). The highest nominal stress range per block is 89 MPa (mold wall), below the 243 MPa allowed for a FAT 71 welded detail at 50,000 cycles. Wear of bushes and mold faces cannot be estimated without soil data, so **R8 is not verifiable at TRL 3.**

Safety features (R12). If the lever is released at the end of the stroke, the soil rebound and column stretch return about **47 J** to it (roughly a 5 kg mass dropped 1 m), so it kicks back upward. The end pawl and the rest catch, spring pawls on the lever bracket, hold the crank pin at the end and at the start of the stroke; the end pawl takes the kickback end on. The T-handle stays 846 mm above the ground at the stop, and the grip is more than 1.5 m from the frame. The crank, connecting link and knee form a scissor point, covered by the linkage guard (BOM item 13). **R12 is met on paper.**

■ 7. Ejection (R2, R5)

After pressing, the lever is returned to its rest stop, which lets the toggle fold under the weight of the piston while the block stays in the mold. The lid is opened and the lever is moved to the eject socket, a seesaw pivoted at $x = 160$ mm, $z = 607.5$ mm on the +X side. Its central arm, with a round nose 160 mm from the pivot, lifts the push rod by its foot, while the 200 mm slot in the rod lets it rise past the toggle pin.

Breaking the block free takes 4.64 kN of wall friction, 4.84 kN with the piston and block weight. The eject lever ratio is 10.3:1 or better (the nose is 139 to 160 mm from the pivot, measured across), so the **grip force is 470 N** or less, within R5 for one operator, over a 60° arc for the 160 mm lift (60 mm up to the block, 90 mm out, 10 mm clear). The 20 x 40 mm arm reaches 145 MPa in bending at the hub (SF 1.72).

■ 8. Mass (R7)

Table 6. Piece masses from the model at 7,850 kg/m³.

Piece	Mass
Base frame with lever bracket, ties and eject posts	45.0 kg
Press core, columns and base beam	28.2 kg
Mold box with hinge and latch lugs	30.4 kg
Lid with ribs, hinge and latch pins	27.0 kg
Piston and slotted push rod	12.6 kg
Toggle links, pins and spacers	24.6 kg
Lever hub, crank, shaft and crank pin	8.2 kg
Lever pipe and T-handle (pipe alone 9.2 kg)	10.4 kg
Eject seesaw, pivot pin, end pawl and rest catch	7.0 kg
Linkage guard (expanded metal at 35 % of solid sheet)	5.5 kg
Bolts	1.2 kg
Total	200.0 kg

R7 is not met on paper: the total is 200.0 kg against the 190 kg accepted by Amish on 2026-09-25 (EPR-DDR-002), 10 kg over. Of the 13 kg added since v0.2, 6.7 kg is the guard and bolts, which v0.2 did not count, and the rest is the round link ends, lugs, larger base plate and other parts added to make the press buildable (EPR-DDR-003). The heaviest piece is 45.0 kg, so the 50 kg piece limit is met. The press is 1,280 x 580 mm in plan, so it fits a 1.5 m pickup bed with the lever removed. Relaxing the total to 200 kg is open decision 1 in EPR-DEC-001.

■ 9. Output and crew (R4)

Table 7. Cycle time (assumed).

Step	Time
Weigh and fill	25 s
Strike, close and latch	8 s
Press, two on the lever	10 s
Return lever, unlatch, open lid	7 s
Move lever to the eject socket	6 s
Eject	6 s
Lift off, gauge, carry to the stack	15 s
Return lever, brush the mold	8 s
Total	85 s

In 7 working hours the press makes **296 blocks per day**, 1 % short of the 300 in R4, so **R4 is at risk**. Two people at the press share the lever and block handling; two others dig, sieve and mix 2,375 kg of soil a day, which takes 9.8 of their 16 person-hours.

■ 10. Material flow, house and carbon (R10, R11)

Per 100 blocks the crew digs 778 kg of soil, sieves out 117 kg of oversize, adds 33 kg of cement and 69 kg of water, and mixes 787 kg including 23 kg of remixed spill and rejects. The press turns out 764 kg of moist blocks, which cure to 712 kg after losing 51 kg of water (Figure 3 in EPR-PRC-001).

A small house with a 22 m perimeter, 2.7 m walls and 15 % openings has 50.5 m² of wall. At 33.3 blocks per m² with 10 mm joints it needs **1,683 blocks**, about 5.7 press days (8.9 m² of wall a day), and 556 kg of cement (11.1 bags of 50 kg). Its 7.07 m³ of walling embodies about **0.35 t CO₂ as CSEB against 4.55 t as fired brick**, 7.6 % of the brick figure (**R11 met on paper**; mortar and transport excluded). R10 (strength with 8 % cement or less) is set by design at 5 % but cannot be verified without block tests.

■ 11. Cost (R13)

The priced BOM totals **\$520**, \$20 over the \$500 value-engineering target in `project.yaml` (`budget_usd`, a hypothetical control target), so **R13 is over the target by USD 20** on indicative prices. The steel alone (200 kg at about \$1.30/kg) is about \$260. The parts added to make the press buildable (EPR-DDR-003) account for the rise from \$488. `budget_usd` is unchanged; the savings worth trying are in the Value engineering section of EPR-DEC-001.

■ 12. Results against requirements

Table 8. Every requirement in EPR-REQ-001 against this note.

ID	Requirement	Value	Target	Status
R1	Block dimensions	Cavity 290 x 140 mm; height 90.0 mm at the stop; wall bulge 0.08 mm	± 2 mm plan, 90 ± 3 mm height	Met on paper, within the fill window
R2	Compaction pressure	2.00 MPa with two operators and fill 0 to +2.7 %; 0.53 MPa with one at 500 N	2.0 MPa or more	At risk
R3	Block strength	Depends on soil, mix and curing	4 MPa dry, 2 MPa wet; 2.1 MPa minimum	Not verifiable at TRL 3
R4	Output	296 blocks per day, crew of four	300 or more	At risk
R5	Operator force and grip height	Peak 668 N total (334 N each for two) at 1.16 m; grip 0.86 to 1.89 m; eject 470 N	500 N or less per operator; grip 0.8 to 1.9 m; peak at 1.0 m or higher	Met on paper (two operators)
R6	Garage-buildable	Plate, sections, pipe and tube; turned pins, shafts and bushes bought	Stick welder, grinder, drill press	Met (design review)
R7	Movable	200 kg total; heaviest piece 45.0 kg; plan 1.28 x 0.58 m	190 kg total; 50 kg per piece; 1.5 m bed	Not met (10 kg over; piece limit met)
R8	Durability	Stress range 89 MPa against 243 MPa; bushes, grease nipples	50,000 blocks; replaceable bushes	Not verifiable at TRL 3 (wear)
R9	Soil test kit	Field tests, chart and a 10 kg scale for fill weighing	Go or no-go in 1 h, full result in 24 h	Met (design review); accuracy not verifiable at TRL 3
R10	Low stabilizer use	5 % cement, 0.33 kg per block	8 % or less reaching R3	Not verifiable at TRL 3
R11	Embodied carbon	7.6 % of fired brick	25 % or less	Met on paper
R12	Safe operation	Rest catch and end pawl on the crank pin; latch and lid rated 173 kN; guard over the linkage; grip over 1.5 m from the frame	No free fall; latch 1.5 x (121.8 kN); guards or 100 mm	Met on paper
R13	Affordable	\$520	At or below the \$500 value-engineering target	Over the value-engineering target by USD 20
R14	Open and documented	Model, drawing, BOM and this note	Build by an outside welder without the author	Not verifiable at TRL 3

Safety: The press delivers 81 kN at 2 MPa and up to 173 kN if two people pull hard at the stop. Keep hands out of the mold, the linkage and the eject arm while anyone is on the lever. The lever kicks back with about 47 J if released at the end of the stroke; engage the end pawl before letting go. Never open the lid until the lever is back on its rest stop.